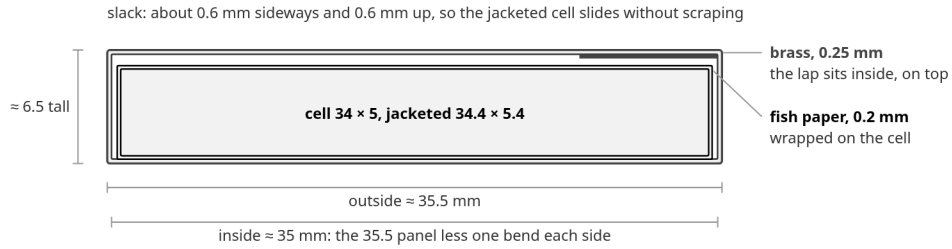
The brass battery sleeve

A flat brass tube, 63 mm long, folded from one piece of 0.25 mm sheet, with the cell inside it in a jacket of fish paper. From outside it is one brass block under the PowerBoost, the same yellow as the rods; the green paper is never seen. One purchase: the sheet. Everything else is on your bench.

Buy: K&S #251 brass sheet, 0.010 in · **Own:** XFJYMXDM fish paper, 0.2 mm · ELEGOO Kapton tape · DIYSELF craft knife · Anezus cutting mat · BAISDY 400 grit · WORKPRO pliers · NEIKO caliper · Harris Stay-Clean SCLF4 · MAIYUM 63/37 solder · 91 % isopropyl alcohol · hot glue gun · Adafruit #258, 1200 mAh

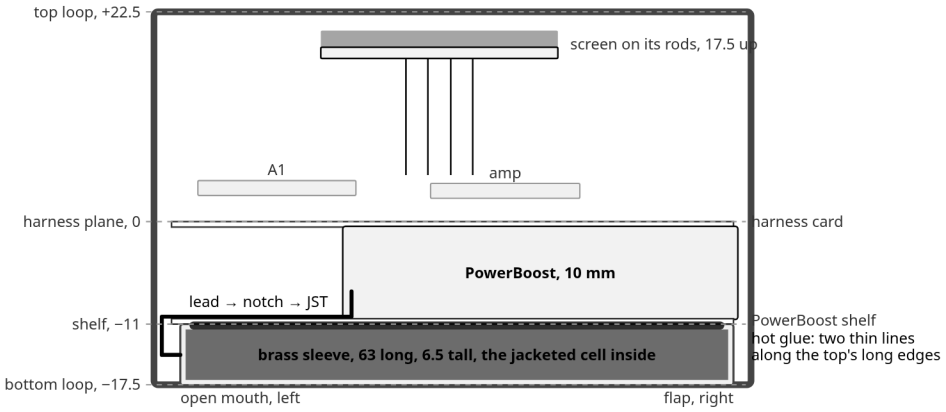
What it is

Looking into the open end: three layers, nothing else



Three layers. Brass is the outside and the structure; fish paper is the insulation and stays wrapped on the cell; the cell floats in the middle with 0.6 mm to spare each way. The brass never touches the cell, the paper never shows.

Where it lives: the bottom storey, hung from the PowerBoost's shelf

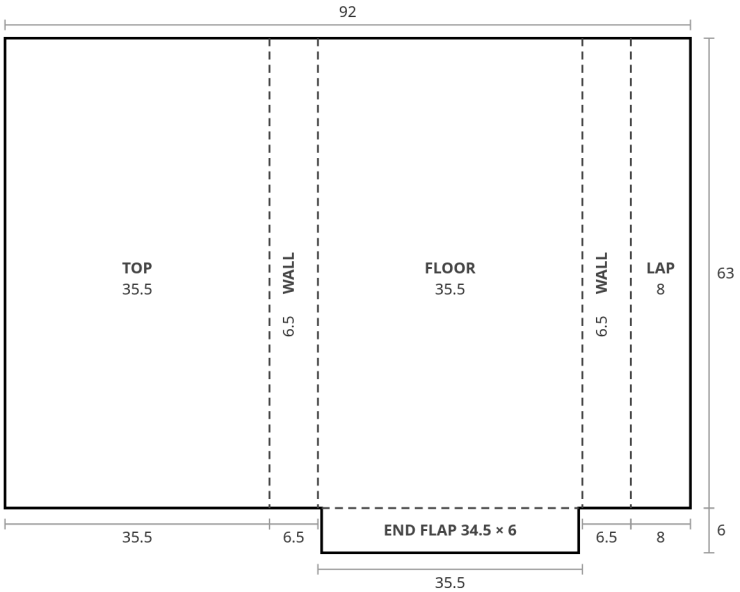


inside 65 × 39 × 40: with the header strips kept, the screen floats at 17.5,
so the battery frame needs 40 inside (43 outside), not the 35 on the whole-build page.
The sleeve is 0.3 mm taller than the cradle it replaces; the shelf rides that much higher.

It replaces the cardstock cradle of the whole-build page, in the same 6.5 mm storey, hung from the underside of the PowerBoost's shelf by two lines of hot glue. The open mouth faces left, under the PowerBoost's JST socket; the lead comes out of it and climbs through a notch in the shelf. In the silver rebuild the sleeve can be soldered straight onto the bottom loop instead; this version comes off with a fingernail. One consequence of keeping the header strips shows here: the screen now floats 17.5 mm up, so the battery frame needs 40 mm inside (43 outside), 5 more than the whole-build page drew.

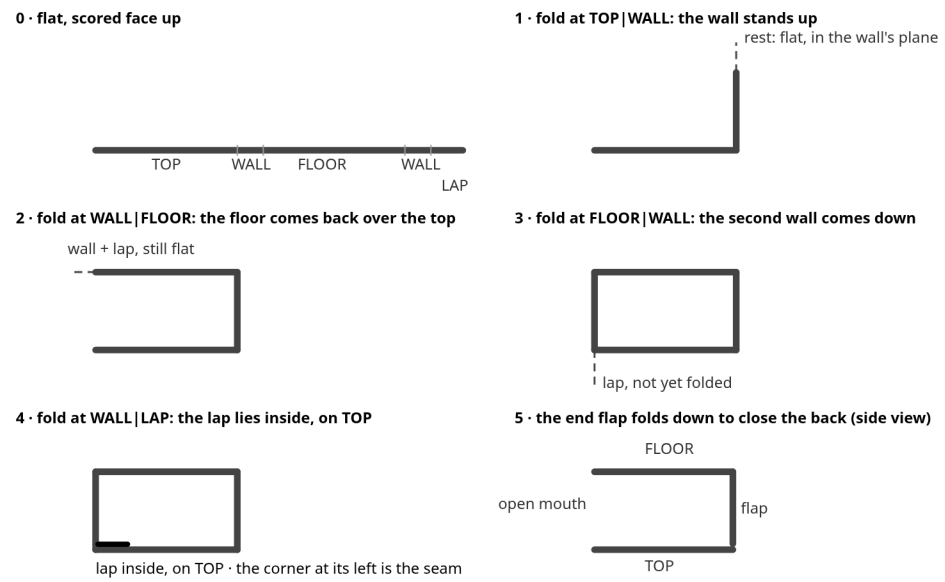
The pattern

The flat pattern: 92 × 69 mm from the brass sheet

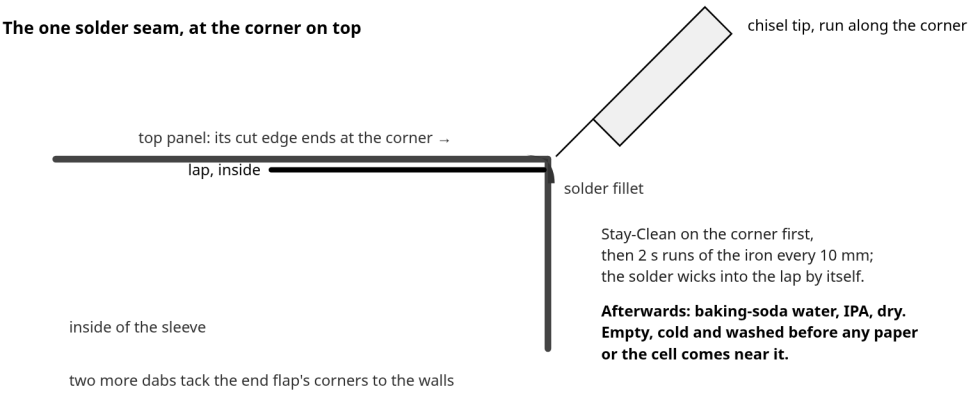


solid = cut · dashed = fold line, scored lightly on this face · this face becomes the inside
the open mouth is the end without the flap; the lead comes out there

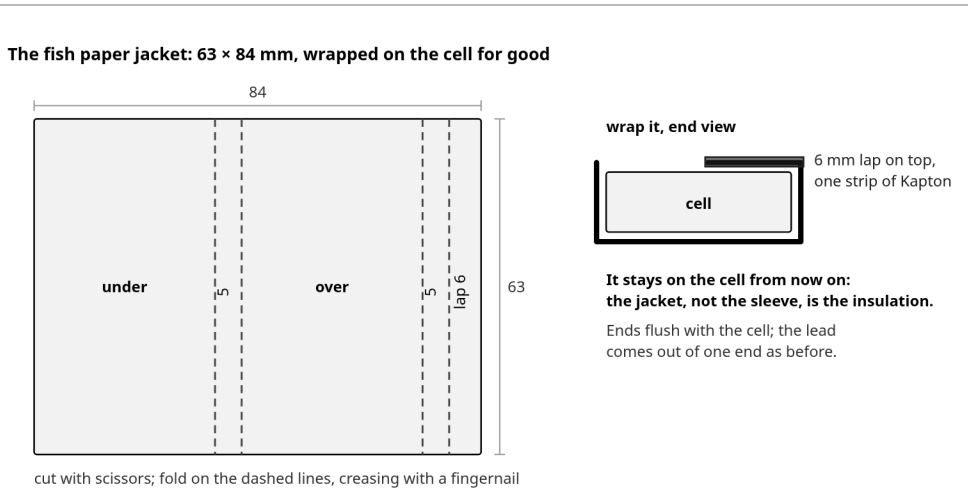
92 × 69 mm, five panels and a flap. TOP and FLOOR are 35.5, the two WALLs 6.5, the LAP 8; one sharp bend eats about 0.45 mm, so the tube comes out ≈ 35 × 6 inside, exactly what a jacketed cell (34.4 × 5.4) needs. The flap on the floor panel folds up to close the back. The 1:1 pattern prints on Letter at Actual size with a 50 mm bar to check it.



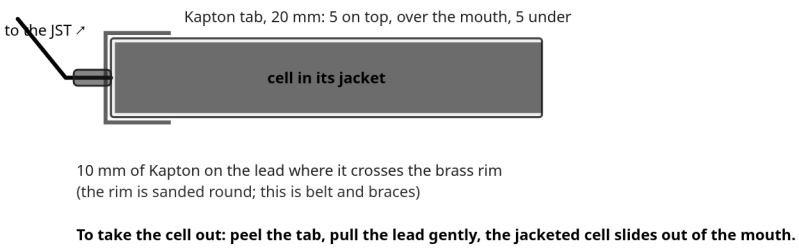
Four folds, all the same way. Keep TOP flat on the table and fold everything else around it: wall up, floor across, wall down, lap in. Because the lap folds inside, the top's outer face stays perfectly flat for the glue, and the seam is the corner line on top, hidden under the shelf.



One seam, soldered from outside. Brass to brass, so this is a Stay-Clean job like the frame: flux along the corner, a chisel tip run along it, the solder wicks into the lap. Two dabs tack the flap's corners. Then neutralise, rinse, dry: nothing comes near the paper or the cell until the sleeve is cold, clean and dry. If you would rather not flux, a strip of Kapton along the inside of the lap holds the tube shut too; it is hidden either way.



The jacket, 84 × 63 of fish paper. Under, up, over, down, 6 mm of lap on top closed with one strip of Kapton. It goes on the cell once and stays: the cell is never handled bare again, and whatever happens to the brass, there is 0.2 mm of flame-rated paper between it and the pouch.



Two pieces of Kapton finish it. A 20 mm tab folded over the open mouth is the seat belt; 10 mm wrapped on the lead where it crosses the brass rim means no edge ever rubs the wire. Peel the tab and the cell slides out, jacket and all.

Build it: 14 steps

Bench rule for this part: the cell stays in its bag until step 11. Brass dust, the knife, the iron and the acid flux are all done by then.

- [ ] **Print the 1:1 pattern at Actual size and measure its 50 mm bar with the caliper.**

  CHECK · 49–51 mm. Outside that, print again with scaling off.

- [ ] **Stick the printed pattern to the brass sheet with a glue stick or four corner dabs of Kapton, near one corner of the sheet so the rest stays whole.**

  CHECK · The pattern lies flat with no bubbles under the fold lines.

- [ ] **Cut along the solid lines with sturdy household scissors, through paper and brass together; 0.25 mm brass cuts like thick card. Cut the two notches beside the flap last.**

  CHECK · One piece, 92 × 69 with the flap; measure the long edge: 91–93.

- [ ] **Score the five dashed lines through the paper with the craft knife against a steel ruler on the cutting mat: two light passes each, a groove you can feel, never a cut.**

  CHECK · Hold it to the light: a bright line on every fold, no crack.

- [ ] **Peel the paper off, then take every edge with the 400 grit, four strokes each, and wipe the dust away with a tissue. Safety glasses on from here to step 8.**

  CHECK · Run a fingertip along every edge: nothing catches.

- [ ] **Fold 1: clamp the sheet under the ruler exactly on the first score line (TOP · WALL), with the scored face up, and lift the overhang to 90° with the flat of the pliers along its whole length.**

  CHECK · A crisp, straight corner; the caliper reads 35.5 outside on the TOP panel.

- [ ] **Folds 2, 3 and 4 the same way, in order: FLOOR across, second WALL down, LAP in until it lies flat on the inside of TOP; then the end flap up to close the back.**

  CHECK · Inside width 35 ± 0.3, inside height 6 ± 0.3 at both ends; a 34.5 × 5.5 cardstock stack slides through without scraping.

- [ ] **Solder the seam: Stay-Clean along the corner line on top, the chisel tip run along it in 2 s passes, solder fed every 10 mm; two dabs on the flap's corners. Then wash: baking-soda water, rinse with IPA, dry an hour.**

  CHECK · The lap does not spring open when you squeeze the walls; no white residue anywhere; dry and cold.

- [ ] **Sand the two mouths round once more, especially the open one, and wipe. The brass is finished.**

  CHECK · No burr at either mouth. If you want it to stay bright, one coat of clear lacquer now, outside only.

- [ ] **Hot-glue it to the underside of the PowerBoost's shelf: two thin lines along TOP's long edges, press for twenty seconds, open mouth at the shelf's left edge where the notch for the lead is.**

  CHECK · Lift the shelf: the sleeve hangs without a wobble, flush with the shelf's left edge.

- [ ] **Cut the jacket, 84 × 63, from the fish paper; wrap the cell under-up-over-down with the lap on top, and close the lap with one strip of Kapton the full length.**

  CHECK · Snug, ends flush with the cell, the lead free at one end; it does not slide on the cell.

- [ ] **Slide the jacketed cell into the sleeve lead-end last, until it stops on the flap; the lead hangs out of the open mouth.**

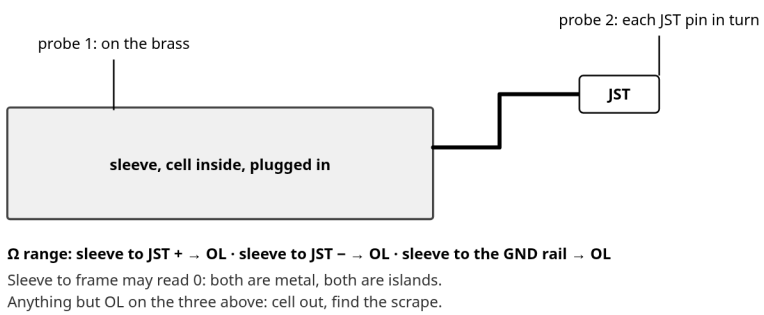
  CHECK · Flush at the mouth, no scrape on the way in; if it drags, the jacket's lap is too thick: re-tape it flatter.

- [ ] **Wrap 10 mm of Kapton on the lead where it crosses the rim, fold the 20 mm tab over the mouth, and feed the lead up through the shelf's notch to the JST (not plugged yet).**

  CHECK · The cell cannot move in any direction by hand.

- [ ] **Meter, Ω range: sleeve to JST +, sleeve to JST −, sleeve to the GND rail. Then plug in with the switch off.**

  CHECK · OL, OL, OL. Anything else: cell out, find the scrape in the jacket.



The last check. The sleeve is allowed to touch the frame (both are metal islands) but nothing electrical. Three OL readings and it is a battery device.

Materials: do you have it all?

Everything but the sheet (the PowerBoost arrived on 2026-10-14 as a third-party copy; its pins still need checking on the bench). Two household items are not in the inventory because they are not purchases: sturdy scissors and a steel ruler (the caliper's beam works for the short lines).

ITEM	STATUS	THE PRODUCT	USED FOR
Brass sheet 0.010 in × 4 × 10 in	**BUY**	K&S #251 brass sheet, 0.010 in	one sheet makes ten sleeves; 0.25 mm folds by hand and holds a crease
Fish paper 0.2 mm	**own**	XFJYMXDM fish paper, 0.2 mm	the jacket, 84 × 63
Kapton tape	**own**	ELEGOO Kapton tape	jacket lap, mouth tab, lead wrap, holding the pattern
Craft knife	**own**	DIYSELF craft knife	scoring the fold lines
Cutting mat	**own**	Anezus cutting mat	under the scoring
400 grit	**own**	BAISDY 400 grit	every edge and both mouths
Chain-nose pliers	**own**	WORKPRO pliers	the folds, flat jaw along the line
Caliper	**own**	NEIKO caliper	the print check and the inside size
Harris Stay-Clean flux	**own**	Harris Stay-Clean SCLF4	the one brass seam; baking soda and IPA afterwards
63/37 solder	**own**	MAIYUM 63/37 solder	the seam
91 % IPA	**own**	91 % isopropyl alcohol	rinse after the acid flux
Soldering station, chisel tip	**own**	—	owned · no link recorded
Hot glue gun, low-temp, white sticks	**own**	—	owned · no link recorded; two thin lines
Cell, 1200 mAh #258	**own**	Adafruit #258, 1200 mAh	62 × 34 × 5; the only cell this fits
Baking soda · safety glasses	**own**	—	owned, bought locally / Amazon #5
Scissors · steel ruler	**own**	—	household; tin snips are not needed for 0.25 mm

PowerBoost 1000C (copy, Amazon #20)	**own**	PowerBoost 1000C copy (Amazon #20)	not part of this piece, but the shelf it hangs from is the PowerBoost's: see the power page

For the nickel-silver rebuild the same pattern works in nickel-silver sheet (not sourced yet); a brass sleeve under a silver frame is also fine.

Sources and limits. Cell size from `docs/INVENTORY.md` (Adafruit #258, 62 × 34 × 5); sheet thickness from the K&S listing; storey heights from `gen/whole_build_page.py`. The bend allowance (0.45 mm per sharp fold in 0.25 mm sheet) is a hand-bending estimate, which is why step 7 measures the inside before anything goes in. Nothing here has been built yet; the numbers are the design.